

Problem C. Experiment with Juice

Time Limit 4000 ms

Mem Limit 262144 kB

OS Windows

Sasha loves physics. Sasha has no friends. Whether these facts are connected or not, Sasha is very upset about his lack of friends. So he tried to impress Vadim with an experiment. Sasha lives on a plane with a Cartesian coordinate system. He took a convex polygon with n vertices numbered from 1 to n , where the i -th vertex has coordinates (x_i, y_i) , and poured juice inside the polygon, filling it to the brim. Then he brought it to Vadim to demonstrate an experiment. Turns out, Vadim doesn't like physics and doesn't care about Sasha's experiments. So he poked a hole in one of the sides of the polygon (but not in one of its vertices!), the hole is a point with coordinates (x_0, y_0) . All the excess juice poured out of the polygon due to the gravity (gravity is pointed in the direction of decrease of coordinate y). But Vadim wasn't satisfied yet, so he started rotating the polygon slowly by angles α_i around the center of the coordinate system, spilling even more juice. You witnessed it all, but no one could stop Vadim. At least you can reassure Sasha by telling him how much exactly juice remained before and after every rotation. We hope you are a good person, and you'll succeed at calming Sasha down.

Input

The first line contains one integer n — the amount of vertices ($3 \leq n \leq 10^5$).

Next n lines contain two numbers x_i, y_i each, separated with a space — coordinates of vertices in counter-clockwise order ($-10^5 \leq x_i, y_i \leq 10^5$). Coordinates are given with exactly three digits after the decimal point.

The $(n+2)$ -th line contain two numbers x_0, y_0 each, separated with a space — coordinates of the hole ($-10^5 \leq x_0, y_0 \leq 10^5$). Coordinates are given with exactly three digits after the decimal point.

The $(n+3)$ -th line contains an integer q — the amount of rotations ($0 \leq q \leq 10^5$).

The last q lines contain one number α_i each — angles of rotations in the order they were performed, counter-clockwise, in degrees ($-360 \leq \alpha_i \leq 360$, rotation by $-x$ counter-clockwise is the same as rotation by x clockwise). The angles are given with exactly three digits after the decimal point.

It is guaranteed that the polygon is convex, no three vertices lay on the same line, and that the hole is located exactly on the polygon's side, but not in one of the vertices.

Output

Output $q + 1$ lines. The first line should contain the area of juice remaining after Vadim poked the hole. Each of next q lines should contain the remaining area of juice after each rotation.

The answer is considered correct if absolute or relative error of each number doesn't exceed 10^{-4} .

Sample

Input	Output
3 -2.000 1.000 1.000 -2.000 4.000 1.000 1.000 1.000	9.0000000000 6.7500000000 4.5000000000 2.2500000000 0.0000000000
4 45.000 45.000 45.000 45.000	

Notes

Illustration of the example

